Ayush Gaggar

(510) 766-8112 | agaggar@u.northwestern.edu | LinkedIn | Website | Github | Scholar

Education

Northwestern University **2022 - 2027**
Ph.D. Candidate in Mechanical Engineering (Robotics) *Chicago, IL*

University of California, San Diego **2020 - 2022**
B.S. Mechanical Engineering, Minor in Entrepreneurship & Innovation, GPA: 3.96/4.00 *La Jolla, CA*

Research & Industry Experience

Ph.D. Research Intern, Multimodal AI **May 2026 - Aug. 2026**
Dolby Laboratories, Inc. *San Francisco, CA*

- Adapted Meta's SAM-Audio foundation model to detect off-screen sound sources in egocentric video, a key step toward closed-loop perception where an agent turns toward what it hears but cannot see.
- Designed InterMaskAttention, a cross-attention layer that enables joint decomposition into per-object stems and a single off-screen residual, improving zero-shot off-screen detection precision by ∼65% over LoRA fine-tuning.
- Built the full pipeline end-to-end, including SAM2 auto-masking, a synthetic off-screen dataset from AVSBench, and LoRA fine-tuning, doubling separation quality (CLAP) over frozen SAM-Audio.

Ph.D. Research Intern, Cooperative Cognition Science **Feb. 2026 - May 2026**
Student Associate at Honda Research Institute, Inc. (HRI) *San Jose, CA*

- Developed a framework for reasoning over the joint distribution of multiple agents' futures, coupling each agent's trajectory distribution via unbalanced optimal transport (OT) within closed-loop model predictive control (MPC).
- Demonstrated on two-agent PushT co-manipulation that OT coordination recovers from local minima, including emergent pivoting behavior, with ∼20% higher success and ∼15% faster completion than independent control.
- Researched a formal measure of when coordination is required, independent of environment encoding task success.

Graduate Researcher, Northwestern University **Jan. 2023 - Present**
Interactive & Emergent Autonomy Lab — PI: Prof. Todd Murphey *Evanston, IL*

- Investigating autonomous data collection to improve latent world model learning for robotic skill acquisition.
- Studying robot decision-making from early latent representations, treating predictive uncertainty as a signal for robust planning rather than a quantity to minimize.
- Previous work includes low-data NeRFs & Gaussian Splatting, linear-complexity ergodic control, dual-arm manipulation, and benchmarking on the Franka Emika arm.
- Technical focus: robot perception, world models, diffusion models, VLMs/VLAs, runtime learning, and embodied AI.

Undergraduate Researcher, University of California, San Diego **Jan. 2021 - Jun. 2022**
Bioinspired Robotics Design Lab — PI: Prof. Mike Tolley *La Jolla, CA*

- Fabricated a squid-inspired underwater robot, evaluating four differential body geometries for locomotion.
- Designed and evaluated five cam profiles using MATLAB-based power and thrust analysis, validating propulsion performance through Arduino-controlled underwater displacement and velocity experiments.

Mechanical/Thermal Engineering Intern **Jun. 2021 - Sep. 2022**
Advanced Micro Devices, Inc. (AMD) *Sunnyvale, CA*

- Designed a test fixture to verify EPYC servers socket pin connectivity under static loads up to 50 kgf.
- Developed a retimer chip cooling assembly deployed in 200+ servers, including BOM and manufacturing documentation.
- Prototyped socket handling trays that accelerated the Genoa launch while preventing damage to up to 50 server sockets.

Publications

[1] **Ayush Gaggar**, Jared Berry, and Todd Murphey. "Does Your World Model Know Actions Have Consequences?" *Workshop at Conference on Robot Learning*, 2026.

[2] Max Muchen Sun, **Ayush Gaggar**, Peter Trautman, and Todd Murphey. "Fast Ergodic Search with Kernel Functions." *IEEE Transactions on Robotics*, 2025.

[3] **Ayush Gaggar**, Todd Murphey. "Data Augmentation for NeRF in the Low Data Limit." *International Conference on Robotics and Automation (ICRA)*, 2025.

[4] Patent: Max Muchen Sun, **Ayush Gaggar**, Peter Trautman, and Todd Murphey. "Kernel-Based Ergodic Search." SN: 18/914929. 2024.

Technical Skills

Embodied AI: PyTorch, JAX, Latent World Models, Diffusion Models, NeRFs, Gaussian Splatting, GPs
Robotics: ROS/ROS2, MuJoCo, Gazebo, PyBullet, Optimal Control, Franka Emika, HDT Adroit, AgileX Piper
Engineering: Python, C++, MATLAB, Linux, Git, SolidWorks, Fusion 360 (CAD/CAM), Simulink, LTSpice, CNC, FDM 3D Printing, Soldering, Instron

Selected Engineering Projects

EKF SLAM for Differential Drive Robot **Jan. 2023 - Mar. 2023**

- Implemented Extended Kalman Filter SLAM from scratch in ROS2 C++, using LiDAR data to localize a Turtlebot3.
- Integrated odometry readings, supervised and unsupervised learning from LiDAR data, and created a simulation environment in RViZ2 for the full functioning SLAM algorithm.

Robot Play: Don't Let the Balloon Touch the Ground **Nov. 2022 - Dec. 2022**

- Controlled a 7DOF Franka arm to track, predict, move to, and hit a balloon such that it would never hit the ground.
- Developed a ROS2 control system from scratch, including forward and inverse kinematics in cartesian and joint space.
- Employed a RealSense RGB-D camera to isolate the centroid of moving, red objects and forward predict its motion.

360° Scanning Sound Array **Jan. 2022 - Jun. 2022**

- Designed a novel testing apparatus to measure noise (dB and freq.) emitted by small-scale UAVs for industry sponsor ATA Inc. as a function of drone's 3DOF moment in 3D space, primarily for national defense interests.
- Primarily manufactured using manual and CNC mills and lathes, with FEA analysis for a 4.24 factor of safety.
- Implemented open loop motor control to determine microphones' spherical positioning with an accuracy of $\pm 0.015°$.

Awards

Doctoral Consortium, *International Conference on Robotics & Automation* **May 2025**
Murphy Fellowship, *Northwestern University* **Sep. 2022**
Magna Cum Laude, *University of California, San Diego* **Jun. 2022**
Best Student Presenter, *Project in a Box, UCSD Undergraduate Conference* **May 2021**
2x Dean's and President's Scholar, *San José State University* **May 2019**
Eagle Scout, *Boy Scouts of America* **Aug. 2018**

Leadership & Services

Teaching Assistant **Fall 2024, Spring 2025**
Northwestern University, Department of Mechanical Engineering *Evanston, IL*

- Assisted in an upper-division Dynamics class with 80 students each quarter with grading and office hours.
- Guest lectured on dynamics concepts like the Lagrangian and Special Rotation matrices.
- Overwhelmingly positive student evaluation (96%) of skill in communication and engagement.

Private Tutor & Peer Educator **2017 - 2025**
UCSD Dept. of Engineering & Freelance *Multiple Locations*

- Mentored 125+ engineering students in mathematics and physics through office hours, review sessions, and independently developed instructional materials.

Museum Presenter - Volunteer **2023 - 2025**
Museum of Science & Industry *Chicago, IL*

- Presented my robotics research during national robotics week, where thousands of children from grades K-12, along with adults, interacted with the experimental systems.